Finite Groups: G-RNP Implies RNP

Automated Math Discovery Smoke Test

June 14, 2026

Source and Open Question

This packet concerns Problem 8.2 from:

> Sheldon Dantas, Michal Doucha, Mingu Jung, Tomas Raunig, *Group equivariant Radon-Nikodym Property and its characterizations*, arXiv:2510.23100.

The source paper asks whether all dotted implications in its summary diagram hold. One dotted implication is

$$G\text{-RNP} \Longrightarrow \text{RNP}.$$

The exact source location is page 34 of the paper.

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Problem 8.2. Decide whether the implications of the dotted arrows hold true.

One of the most mysterious implications is the one between the classical Bishop-Phelps property and the G-Bishop-Phelps property. Notice that it is the only instance of the phenomenon, among the properties we have considered here, that the classical property does not obviously imply its G-analogue. It is possibly also related to the apparent difficulty of showing that strong G-dentability implies the G-BP (for a compact group G). It would be certainly very interesting to find an example of G-Banach space with the BP which is strongly G-dentable but fails the G-BP, thereby disproving both implications at once.

Definitions Used

Let a topological group G act continuously by linear isometries on a Banach space X. Following the source paper, X has the G-Radon–Nikodym property if for every finite measure space (Ω, Σ, μ) with a continuous action of G by regular set isomorphisms, every G-quasi-invariant X-valued vector measure m on Σ satisfying $|m| \ll \mu$ has a Bochner integrable density.

When the action on the measure space preserves the measure, the G-quasi-invariance condition reduces to

$$m(gA) = g\,m(A) \qquad (g \in G,\ A \in \Sigma).$$

Proof Intuition

For a finite group, one can encode an arbitrary vector measure as an equivariant one by placing its copies on the finitely many sheets $\{g\} \times \Omega$. The group acts freely by permuting the sheets, while the target action twists the copied measure. If the induced equivariant measure has a Radon–Nikodym derivative, then restricting that derivative to any one sheet recovers a derivative for the original vector measure.

Candidate Partial Result

Theorem 1. *Let G be a finite Hausdorff topological group and let X be a G-Banach space. If X has the G-RNP, then X has the classical RNP.*

Proof. Since finite Hausdorff groups are discrete, the action below is continuous. Let (Ω, Σ, μ) be a finite measure space and let $m : \Sigma \to X$ be a countably additive vector measure with $|m| \ll \mu$. It is enough to prove that m has a Bochner integrable density.

Write $n = |G|$ and equip

$$\widetilde{\Omega} = G \times \Omega, \qquad \widetilde{\Sigma} = \mathcal{P}(G) \otimes \Sigma, \qquad \widetilde{\mu} = \frac{1}{n} \sum_{g \in G} \delta_g \otimes \mu$$

with the measure-preserving left-translation action

$$h \cdot (g, \omega) = (hg, \omega).$$

For $E \in \widetilde{\Sigma}$, let

$$E_g = \{\omega \in \Omega : (g, \omega) \in E\}.$$

Define an X-valued vector measure M on $\widetilde{\Sigma}$ by

$$M(E) = \frac{1}{n} \sum_{g \in G} g\, m(E_g).$$

This is countably additive because it is a finite sum of countably additive vector measures. Moreover,

$$|M|\,(E) \leq \frac{1}{n} \sum_{g \in G} |m|\,(E_g),$$

so $|M| \ll \widetilde{\mu}$.

We now check equivariance. For $h \in G$,

$$(hE)_g = E_{h^{-1}g}.$$

Therefore

$$M(hE) = \frac{1}{n} \sum_{g \in G} g\, m(E_{h^{-1}g}) = \frac{1}{n} \sum_{k \in G} hk\, m(E_k) = h\, M(E).$$

Because the action on $\widetilde{\Omega}$ is measure preserving, this says that M is G-quasi-invariant in the sense of the source paper.

By the G-RNP of X, there exists $F \in L_1(\widetilde{\Omega}, \widetilde{\Sigma}, \widetilde{\mu}; X)$ such that

$$M(E) = \int_E F\, d\widetilde{\mu} \qquad (E \in \widetilde{\Sigma}).$$

Fix $g_0 \in G$. For every $A \in \Sigma$,

$$M(\{g_0\} \times A) = \frac{1}{n} g_0\, m(A)$$

while

$$\int_{\{g_0\}\times A} F\, d\widetilde{\mu} = \frac{1}{n} \int_A F(g_0, \omega)\, d\mu(\omega).$$

Thus

$$m(A) = \int_A g_0^{-1} F(g_0, \omega)\, d\mu(\omega) \qquad (A \in \Sigma).$$

The function $\omega \mapsto g_0^{-1} F(g_0, \omega)$ is Bochner integrable because G is finite and $F \in L_1(\widetilde{\mu}; X)$. Hence m has a Radon–Nikodym derivative. Since (Ω, Σ, μ) and m were arbitrary, X has the classical RNP. $\square$

Why This Is Partial

The theorem resolves the dotted implication G-RNP $\Rightarrow$ RNP only when the acting group is finite. The argument uses positive-measure singleton sheets $\{g\} \times \Omega$; it does not directly extend to infinite compact groups, where Haar-measure singletons are null. The general dotted implication from the source paper remains open in this packet.

Verification Notes

The proof is self-contained. The key points for review are:

- the induced measure $M(E) = n^{-1} \sum_g g\, m(E_g)$ has finite variation and is absolutely continuous with respect to $\widetilde{\mu}$;
- the left-translation action makes M equivariant, hence G-quasi-invariant in the measure-preserving case;
- because G is finite, the derivative on one sheet is an honest $L_1(\mu; X)$ function and recovers the original vector measure.

No computational checks were needed.

Novelty and Literature Check

The run's lightweight registry, solution, attempt, and proof-gap indexes were searched for arXiv:2510.23100 and for the phrases "Group equivariant Radon", "G-RNP", "RNP", "dotted arrows", and related dentability/BP keywords. No prior packet or attempt for this paper was found. The source paper itself does not answer the G-RNP $\Rightarrow$ RNP dotted arrow near Problem 8.2. No external literature search was performed for this partial finite-group subcase.

References

- Sheldon Dantas, Michal Doucha, Mingu Jung, Tomas Raunig, *Group equivariant Radon-Nikodym Property and its characterizations*, arXiv:2510.23100.

Human Review Recommendation

Review as a likely valid partial result. It is a small but exact subcase of one dotted implication in the summary diagram, not a full solution of Problem 8.2.